ATP 2026 | ENEC CHALLENGE

MARLIN

Marine Adaptive Response and Lifecycle Intervention Network







A layered, proven response to jellyfish blooms, with less manual work

Post-alert management of seasonal blooms: bubble curtain plus angled boom, adaptive control software and a manual-catch backup

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Two proven layers, adaptive control, a manual-catch Layer 3

1 The problem

Bloom swarms clog cooling-water screens and can trip reactors. Gravelines lost four units in Aug 2025 and three in Aug 2026. After the warning, clearing is still manual, equipment-heavy and sustained.

2 The solution

A seabed bubble curtain lifts drifting jellyfish to the surface and an angled floating boom guides them to one collection point. Control software adapts both to the bloom. Layer 3, a small manual-catch crew, sweeps any straggler; the existing screens stay as the last line.

3 Why it is credible

A flume study found a bubble plume plus floating boom gathers jellyfish for pump removal (Lo, 1996). Bubble curtains cut jellyfish by 77% in Tasmanian pens. Booms have flume-based design guidance for power-station intakes.

Next step Build and test the controller in simulation and a bench flume with density-matched analogues, then a sheltered-water pilot. No change to plant or nuclear-safety systems.

Jellyfish blooms are shutting down coastal plants



Seawater intake channel at a coastal nuclear plant (team reference photo)

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Gravelines, France, Aug 2025: jellyfish in the pump-station filter drums shut four reactors. It happened again in Aug 2026.

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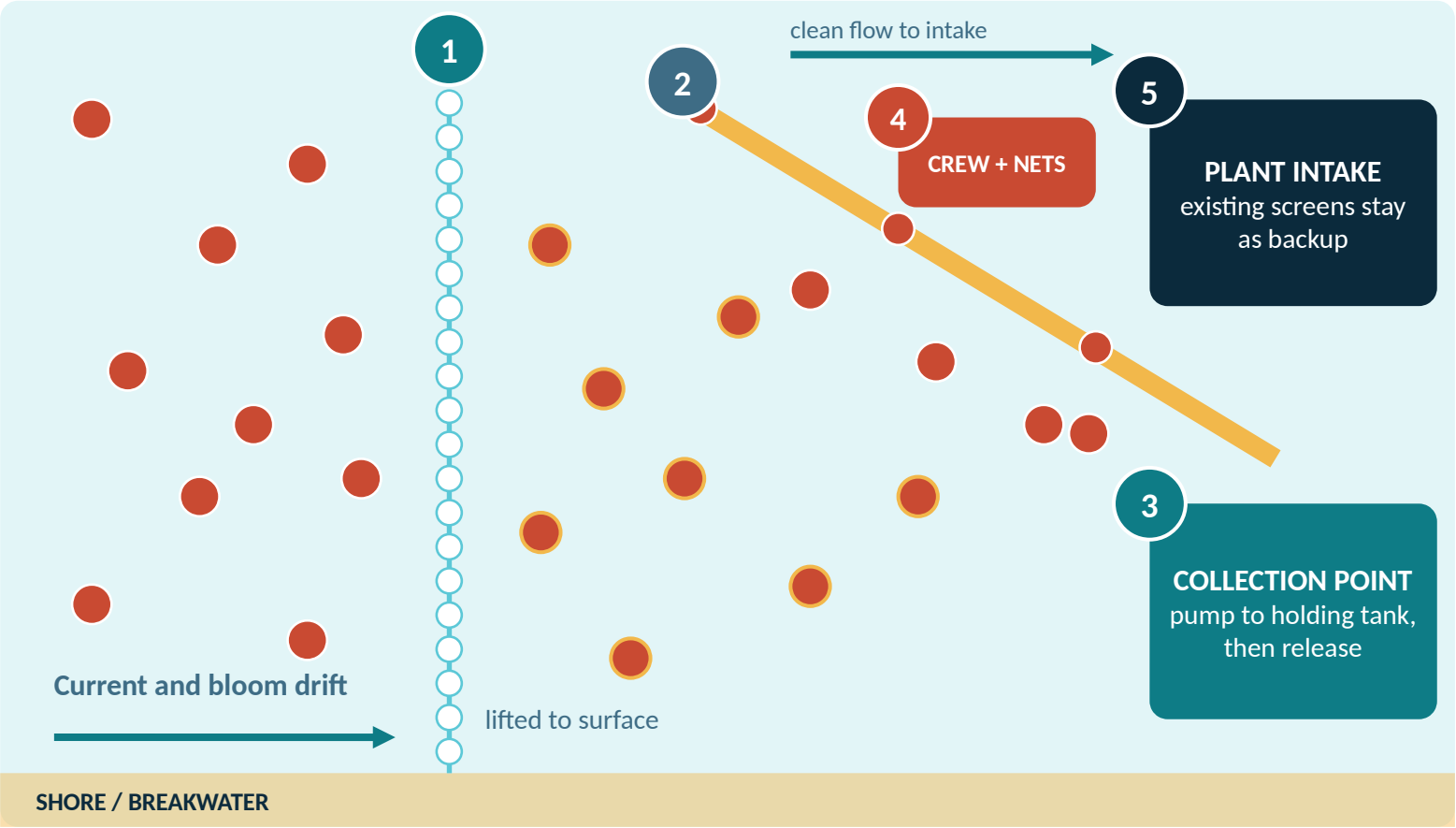
Ghubra desalination, Oman: 300 tonnes of jellyfish damaged intake screens and cut output by 50%.

Gulf-wide

2002-03 Crambionella bloom damaged power-plant cooling systems in the Gulf of Oman and Persian Gulf. Al-Zour, Kuwait, saw screens blocked in 2017.

Target Catostylus mosaicus, 30-45 cm, dense and firm (ENEC brief). The system must handle large, continuous volumes, not single animals.

How it works: lift, guide, collect, back up



Schematic top view, not to scale. Boom angle, curtain length and crew station need site data.

1 **Bubble curtain**

Seabed air hose lifts drifting jellyfish to the surface.

2 **Angled boom**

Guides them along its face to one collection point.

3 **Collection point**

Gentle pump to a holding tank, then supervised release.

4 **Layer 3: manual catch**

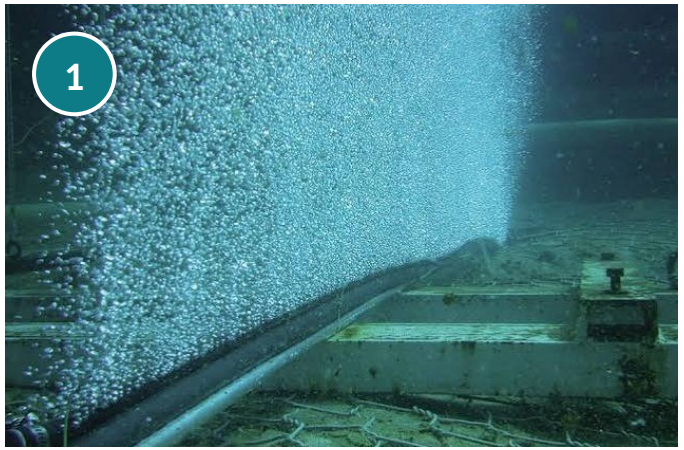
A small crew with dip nets clears any rare straggler.

5 **Intake stays as is**

Existing screens remain the last line of defense.

Control software adapts layers 1 and 2

Each layer is already used in the field



Layer 1 | Bubble curtain

- Tasmania: 77% fewer jellyfish in a 12-pen trial, now rolled out to 160+ pens
- Ireland: an 8 m high-airflow curtain deflected large jellyfish (Haberlin et al., 2021)
- Chile: 70+ salmon sites, plus power-plant inlets (vendor-reported)

Design limit: waves and low airflow let jellyfish through; small fragments pass.



Layer 2 | Angled floating boom

- Flume-based design guidance for debris booms at power-station intakes; bent booms suit the widest flow range
- Flush-faced booms clear themselves when set at an angle to the current (manufacturer data)
- Net-skirt booms already keep jellyfish out of desalination intakes (Elastec)

Design limit: holds only at low current across the boom, so the angle sets the load.

The hybrid is backed by published tests

Claim	Evidence	Strength
A bubble barrier alone is not enough	Lo (1991), flume: a net downstream of the barrier is recommended	**Lab**
Bubbles plus a boom gather jellyfish for removal	Lo (1996), flume: the plume lifted jellyfish to the surface, where they gathered at the boom for pump removal	**Lab**
Bubble curtains cut jellyfish at scale	Tasmania: 77% fewer in 12 pens. Mowi Portnalong: up to 80% fewer inside the curtain	**Field, operator-reported**
Results depend on size, waves and airflow	Haberlin et al. (2021): large jellyfish deflected; waves and low airflow raised pass-through	**Peer-reviewed field and flume**
Angled booms guide debris at intakes	Ettema and Kubit (2000): flume-tested design guidelines for debris booms at power-station intakes	**Published flume study**
Airflow and boom angle are real control knobs	Haberlin et al. (2021): airflow and waves changed how many got through. Adjustable-angle booms exist (US patent 5,695,300)	**Flume plus patent**

Gap we will close No published test of this hybrid on a dense 30-45 cm species at full scale, and none of an adaptive controller for it. Our simulation and flume plan target both.

A software brain that adapts the response

01 | ALERT

ENEC bloom warning

Existing offshore detection starts the loop.

02 | SENSE

Count and track

Camera or sonar estimates arrivals; sensors read current and waves.

03 | DECIDE

Estimate and choose

Software sets air level and boom angle from arrival rate and direction.

04 | ACT

Adjust or stop

Air, boom and collector follow the bloom; all stop when arrivals stop.

If the software sees	Then it
Arrivals rise	Raise air flow to its set-point; re-angle the boom to keep cross-flow within its limit
Arrival direction shifts	Swing the boom toward the new approach
Arrivals near zero for a set time	Stop air, park the boom, keep watching
Waves, boom load or intake pressure too high	Air off or boom retracted; alert the operator

Guardrails

- Local interlocks bound every set-point
- Operators authorize deployment and releases
- Rules first; machine learning only once data exist
- Starts from the ENEC alert; does not replace detection

Built around the known limits and marine life

KNOWN LIMIT	OUR RESPONSE
Waves cut curtain performance	Use the breakwater-sheltered approach; software stops air in rough seas
Small fragments pass through	Boom skirt, manual crew and existing screens act as further lines
Boom face can clog; intake flow may change	Angle lets current clear the face; retract at pressure limit
Dense 30-45 cm bodies may resist lift	Test lift in the flume with density-matched analogues
Sensor counts may be wrong in turbid water	Conservative defaults, operator override, counts validated in the flume

Marine-life safeguards

- Air and floating guides only: no chemicals, no shredding
- Gentle handling aims to release jellyfish alive, away from the intake; survival is measured
- Non-target animals are detected before handling; pause and release by protocol
- Bubble noise and light effects on fish are measured in sheltered-sea tests
- Mass balance recorded: delivered, held, escaped, bycatch

Low-cost, modular, and testable before any pilot

Cost-conscious by design

Passive first: boom needs no power; air runs only while jellyfish arrive

Modular: sections are repaired or extended one at a time

Off-the-shelf: booms, diffuser hose, compressors, pumps

Scales by length: add sections for a wider channel

Layer 3 crew on call: manual catch is a rare backup

Capital and running costs need supplier quotes; none are assumed here.

1

Bench flume (Demo Days)

Scaled curtain and boom with density-matched analogues. Test the controller on changing arrival rates and directions.

2

Sheltered-water trial

Permitted deployment with ecological observation, load cells and intake-pressure logging.

3

ENEC-approved pilot

Operator-led demonstration under approved boundaries, with no change to plant systems.

To prove A clear cut in bloom mass reaching the intake versus no response (our screening model suggests roughly 45-65% in calm seas and about 11% in a 2 m storm, so flume tests must measure the real figure); the controller re-aims and stops correctly; bycatch, pressure change, kWh per tonne and staff-hours recorded.

Assumptions, limits, and sources

KEY ASSUMPTIONS

- Public information only; no ENEC facility or operating data used
- Breakwater-sheltered approach with low waves (to confirm)
- Target: Catostylus mosaicus, 30-45 cm, dense (ENEC brief); lift behavior untested
- Current, depth and bathymetry unknown; boom angle and curtain length follow a site survey
- Aquaculture and lab results do not guarantee performance at a nuclear intake
- Sensor counts and the controller are unvalidated; first built and tuned on simulated and flume data

SOURCES

Lo (1991), IAHR: iahr.org/library/info?pid=6399
Lo (1996), Ocean Engineering 23(6): trid.trb.org/View/480225
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Pulse Tasmania: Tassal bubble-curtain rollout
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Politico (Aug 2025), NEI Magazine (Aug 2026): Gravelines shutdowns
Gulf News: Jellyfish choke Oman desalination plants
J. Mar. Biol. Assoc. UK: Crambionella orsini bloom, 2002-03
PAAET Kuwait: Preventing jellyfish attacks on power plants
Elastec; EcoCoast/Bolina: boom and debris-boom product data
US patent 5,695,300: adjustable-angle recovery boom
ENEC: ATP 2026 challenge statement (challengeon.atrc.ae)

Declaration AI tools supported research and drafting. The team reviewed every claim and owns the proposal.

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